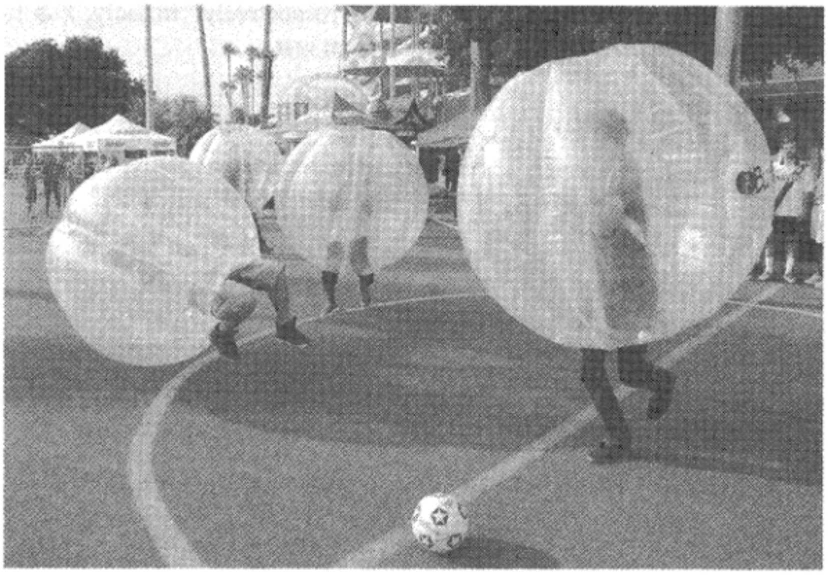
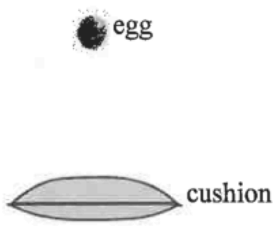
12. Players of "bubble soccer" wear air-filled plastic "bubbles" as shown.



Which of the following statements best explains why the bubble can reduce the chance of injury during a collision ?

A. The bubble increases the mass of the player, thus the momentum of the player increases.
B. The bubble increases the air resistance acting on the player.
C. The bubble lengthens the impact time during a collision.
D. Like a balloon, the bubble provides a lifting force to the player.

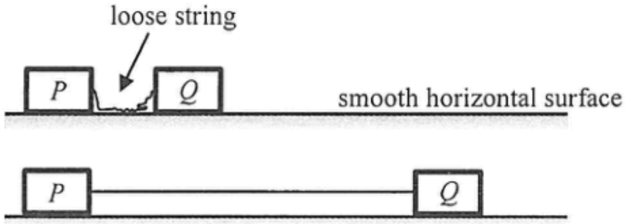
7. An egg is released from a height and falls towards a cushion without breaking.



Which of the following is the most probable explanation ?

A. The cushion makes the egg stop at a shorter distance.
B. The cushion helps lengthen the time of impact.
C. The cushion helps reduce part of the gravitational force acting on the egg.
D. The cushion increases the reaction force acting on the egg during impact.

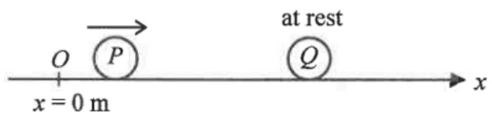
8. On a smooth horizontal surface, two identical blocks P and Q are connected by a light inextensible string. The blocks are at rest and the string is loose initially.

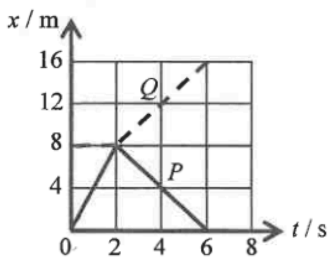


Q is given a speed of 4 m s^{-1} and moves to the right. Find the speeds of the blocks when the string just becomes taut and P starts to move.

	block P	**block Q**
A.	1 m s^{-1}	1 m s^{-1}
B.	2 m s^{-1}	1 m s^{-1}
C.	2 m s^{-1}	2 m s^{-1}
D.	4 m s^{-1}	2 m s^{-1}

12.





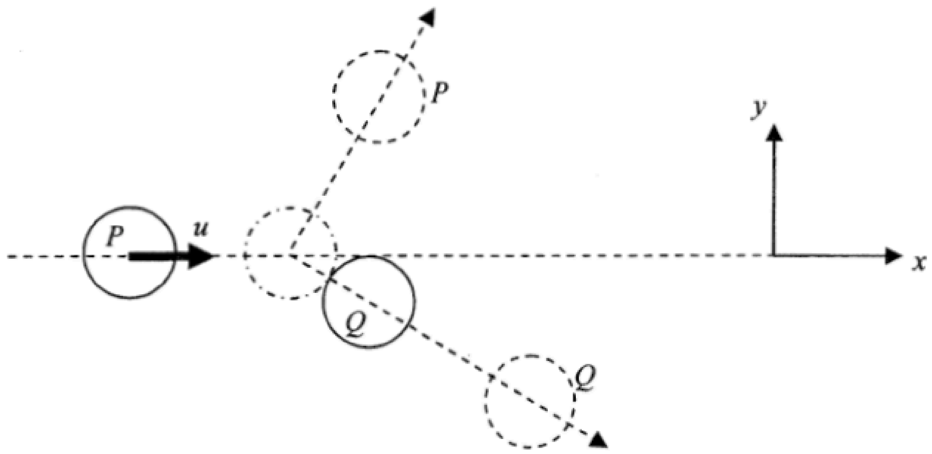
On a smooth horizontal surface, sphere P travelling along the x-axis collides head-on with another sphere Q, which is initially at rest at $x = 8$ m. The graph shows the displacement-time (x-t) relationship of P (solid line) and Q (dotted line). The collision takes place at $t = 2$ s and the collision time is negligible. Which statement is correct ?

A. The collision is perfectly inelastic.
B. The mass of P is larger than that of Q.
C. P still moves along $+x$ direction after collision.
D. P and Q move with the same speed after collision.

12. At a certain instant, an object flying horizontally to the right at 1 m s^{-1} suddenly explodes into two fragments of mass ratio 1 : 2. If just after the explosion the more massive fragment flies to the right at 3 m s^{-1}, what would happen to the other fragment just after the explosion ?

A. It would fly to the left at 3 m s^{-1}.
B. It would fly to the left at 4 m s^{-1}.
C. It would be momentarily at rest.
D. It would fly to the right at 1 m s^{-1}.

13. On a smooth horizontal surface, a circular disk P moving at velocity u along the x direction collides obliquely with an identical disk Q initially at rest as shown below. The mass of each disk is m. Which statements about the collision is/are correct ?



(1) Momentum of the system in the y direction is not conserved.

(2) The total kinetic energy of P and Q after collision is $\frac{1}{2}mu^2$ if the collision is perfectly elastic.

(3) Speed of Q after collision is less than u.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

10.

Two identical spheres are moving in opposite directions with speeds u and v (with $u > v$) respectively as shown. They make a head-on collision. Which of the following diagrams show(s) a *possible* situation of the spheres after collision ?

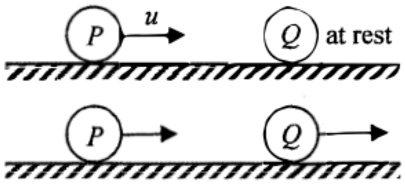
(1) 

(2) 

(3) 

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

12. On a smooth horizontal surface, a marble P moving with speed u collides head-on with another marble Q, which is at rest. After collision, P and Q move with different speeds as shown.



Which of the following statements about this collision is/are correct ?

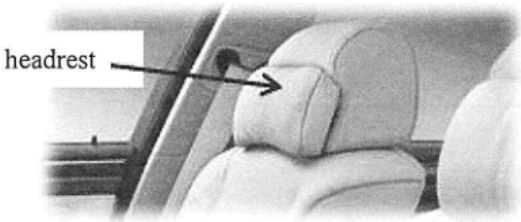
(1) During collision, the force acting on Q by P is equal and opposite to that acting on P by Q.

(2) The total momentum of the two marbles is conserved only when the collision is perfectly elastic.

(3) The kinetic energy lost by P must be equal to that gained by Q.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

5.

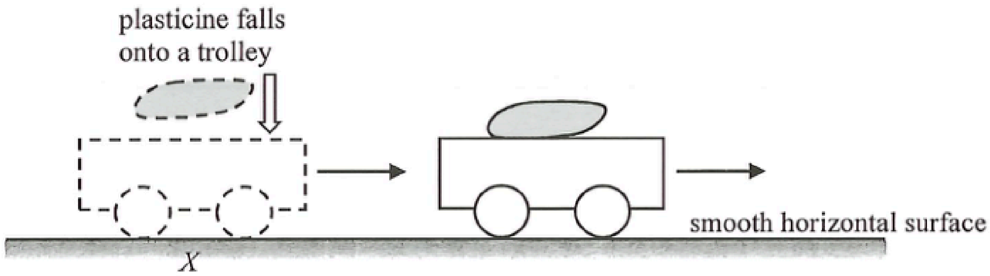


For a car travelling on a highway, which of the following statements about the safety design of the headrest is/are correct ?

(1) As the headrest is soft, it can reduce the force exerted on the passenger's head during impact.
(2) It can minimise injury of the passenger when the car is struck by another one from behind.
(3) It can minimise injury of the passenger when the car brakes suddenly.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

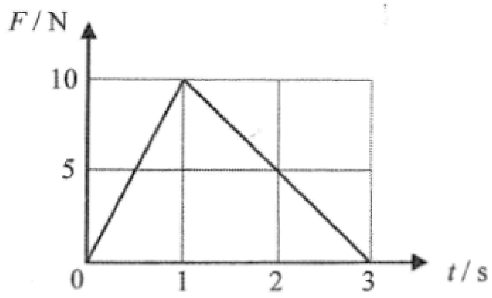
11. A trolley moves with a constant velocity along a smooth horizontal surface. When the trolley reaches point *X*, a plasticine falls vertically onto it. After the collision, they stick together and continue to move forward.



Which description about the **total linear momentum of the trolley and the plasticine** just before and just after collision is correct ?

	along the horizontal direction	along the vertical direction
A.	conserved	conserved
B.	conserved	not conserved
C.	not conserved	conserved
D.	not conserved	not conserved

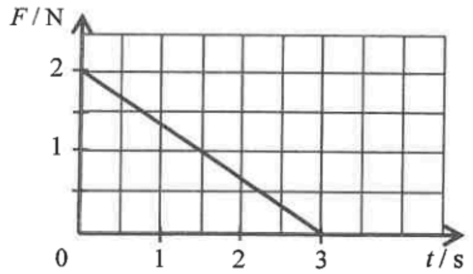
6.



An object of mass 3 kg is initially at rest on a smooth horizontal ground. A force F is applied horizontally to the object such that the magnitude of F varies with time t as shown. What is the speed of the object at $t = 3$ s ? Neglect air resistance.

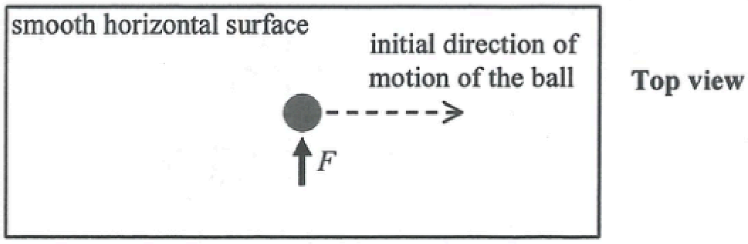
A. 2.5 m s^{-1}
B. 5 m s^{-1}
C. 10 m s^{-1}
D. 15 m s^{-1}

8. An object of mass 2 kg moves with velocity 1 m s^{-1} initially. It is acted upon by a force F which varies with time t as shown in the graph. The force and the object's velocity are in the same direction. Find the speed of the object at $t = 3$ s.



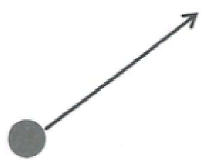
A. 1.5 m s^{-1}
B. 2.5 m s^{-1}
C. 3.5 m s^{-1}
D. 4.5 m s^{-1}

10.



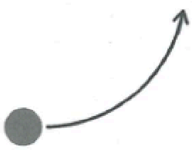
The above figure shows a ball moving with a constant speed along a straight line on a smooth horizontal surface. At a certain instant, the ball is acted on by a force F for a very short time as shown above. Which **subsequent** path below would the ball most closely follow ?

A.

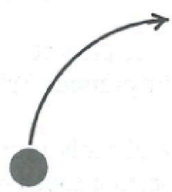


B.

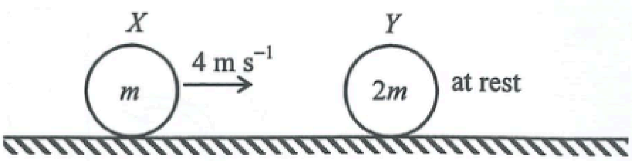
C.



D.



11.

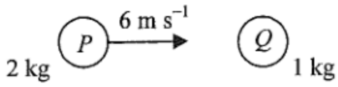


On a smooth horizontal surface, sphere X of mass m travels with speed 4 m s^{-1}. It collides head-on with another sphere Y of mass $2m$, which is at rest initially. Which of the following can be the speed of Y just after collision ?

(1) 1 m s^{-1} (2) 2 m s^{-1} (3) 3 m s^{-1}

A. (1) only
B. (2) only
C. (1) and (2) only
D. (2) and (3) only

7.



A sphere P of mass 2 kg makes a head-on collision with another sphere Q of mass 1 kg which is initially at rest. The speed of P just before collision is 6 m s^{-1}. If the two spheres move in the same direction after collision, which of the following could be the speed(s) of Q just after collision ?

(1) 2 m s^{-1}
(2) 4 m s^{-1}
(3) 6 m s^{-1}

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

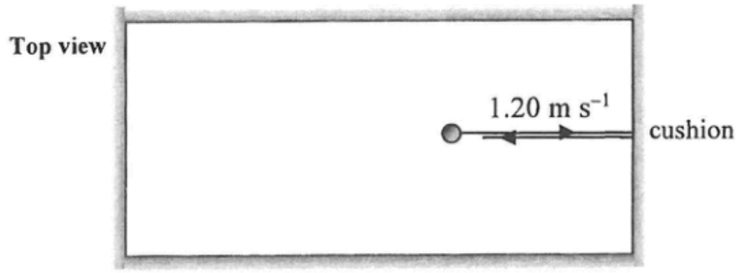
8. A stationary uranium nucleus ${}^{238}_{92}\mathrm{U}$ decays to give a thorium nucleus ${}^{234}_{90}\mathrm{Th}$ and an α particle ${}^{4}_{2}\mathrm{He}$.

$$ {}^{238}_{92}\mathrm{U} \longrightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He} $$

Which of the following correctly describes the situation about the ${}^{234}_{90}\mathrm{Th}$ nucleus and the α particle just after the decay ?

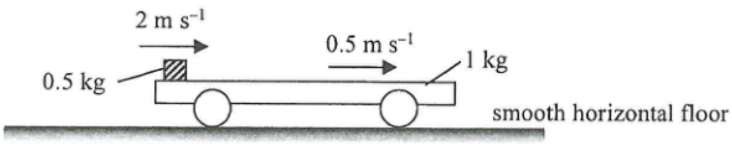
	magnitude of momentum p	kinetic energy KE
A.	$p(\mathrm{Th}) = p(\alpha)$	$\mathrm{KE(Th)} < \mathrm{KE}(\alpha)$
B.	$p(\mathrm{Th}) > p(\alpha)$	$\mathrm{KE(Th)} > \mathrm{KE}(\alpha)$
C.	$p(\mathrm{Th}) = p(\alpha)$	$\mathrm{KE(Th)} > \mathrm{KE}(\alpha)$
D.	$p(\mathrm{Th}) = p(\alpha)$	$\mathrm{KE(Th)} = \mathrm{KE}(\alpha)$

11. On a billiard table, a billiard ball of mass 0.16 kg and moving with a uniform speed of 1.20 m s^{-1} collides elastically with the cushion. It rebounds with the same speed normal to the cushion as shown. If the impact time is 0.0003 s, find the average force acting on the ball by the cushion during collision.



A. 0 N
B. 640 N
C. 1280 N
D. 2560 N

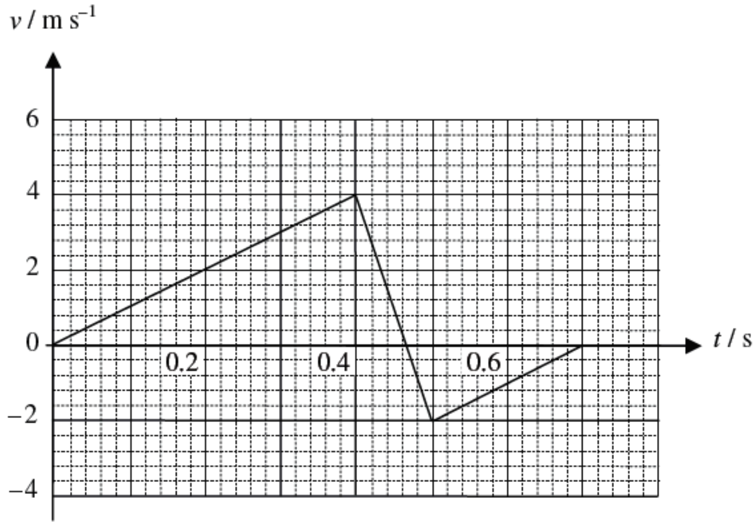
10. A 1-kg trolley with a 0.5-kg block placed on its horizontal platform is resting on a smooth horizontal floor. The block is then given a sharp push to the right. At an instant after the push, the block and the trolley are moving to the right at 2 m s^{-1} and 0.5 m s^{-1} respectively as observed from the floor.



Finally, they travel together with the same speed v. Find v. Neglect air resistance.

A. 0.5 m s^{-1}
B. 1 m s^{-1}
C. 1.5 m s^{-1}
D. 2 m s^{-1}

13.



A ball of mass 0.2 kg is released from rest. It hits the ground and rebounds. The velocity-time graph of the ball is shown above. Which of the following statements are correct ?

(1) The magnitude of the change in momentum of the ball during the collision is 1.2 kg m s^{-1}.

(2) The magnitude of the average force acting on the ball by the ground during the collision is 12 N.

(3) There is mechanical energy loss during the collision.

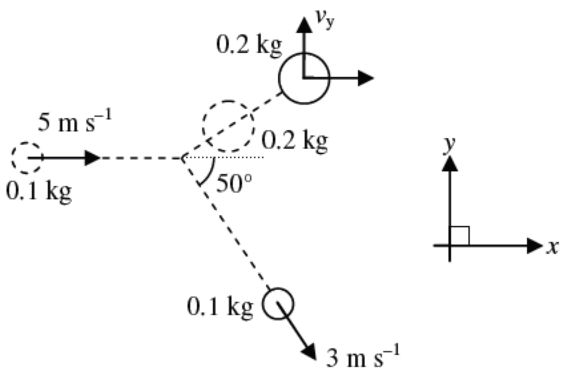
A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

*14. A disc of mass 0.1 kg and velocity 5 m s^{-1} strikes a stationary disc of mass 0.2 kg on a smooth table. After the collision, the 0.1 kg disc moves with a speed of 3 m s^{-1} at 50° to the x direction. Find the component of the velocity of the 0.2 kg disc in y direction, v_y, after the collision.



A. 1.15 m s^{-1}
B. 1.54 m s^{-1}
C. 1.92 m s^{-1}
D. 2.01 m s^{-1}

13. A car *P* of mass 1000 kg moves with a speed of 20 m s^{-1} and makes a head-on collision with a car *Q* of mass 1500 kg, which was moving with a speed of 10 m s^{-1} in the opposite direction before the collision. The two cars stick together after the collision. Find their common velocity immediately after the collision.

A. 2 m s^{-1} along the original direction of *P*
B. 2 m s^{-1} along the original direction of *Q*
C. 14 m s^{-1} along the original direction of *P*
D. 14 m s^{-1} along the original direction of *Q*